

Course: Artificial Intelligence in Software Engineering

Course Code: MCI1014

ASSIGNMENT 1:

AI-Assisted Battery Warranty Verification and Claim Management System for Bateriku.com

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1.0 INTRODUCTION

Bateriku.com provides car battery-related services to customers who may experience vehicle starting problems or other battery-related issues. Car batteries play an important role in ensuring that vehicles operate reliably. When a car battery experiences problems during its warranty period, customers may submit warranty claims to obtain an inspection and, if the claim meets the applicable warranty conditions, the battery will be replaced. An efficient warranty verification and claim management process is therefore important for customer satisfaction, service efficiency, and accurate warranty record management.

Based on the warranty process described for Bateriku.com, customers initiate a warranty claim complain, and an administrator checks the vehicle and warranty information in the system (JBK). If the battery is still within the warranty period, a technician is dispatched to the customer's location to inspect the battery. If a battery problem is identified, the technician provides a temporary replacement, known as a floating battery, for the customer's use. The battery submitted for warranty is then sent to Bateriku.com's headquarters for further inspection. If the battery is confirmed to be defective and the claim satisfies the applicable warranty conditions, the customer receives a replacement battery. According to the described process, the warranty period continues from the original warranty period rather than restarting with a new period.

However, one potential challenge in this process occurs when the administrator cannot find the customer's vehicle or battery record in the system. Missing records may make it difficult to verify the purchase date, warranty validity, battery details, and claim history. This situation may delay the verification process and require additional checks before the claim can proceed. The frequency and impact of this problem should be assessed using actual operational evidence.

2.0 PROBLEM STATEMENT

Bateriku.com provides a battery warranty service that allows customers to submit warranty claims when they experience battery problems. When a complaint is received, the administrator will check the system (JBK) to determine whether the vehicle and battery records exist and whether the warranty is still valid. If the claim proceeds, a technician visits the customer, provides a temporary floating battery when required, and sends the suspected defective battery to headquarters for inspection. If the battery is confirmed defective and the claim is approved, a replacement battery is provided under the original warranty period.

However, some warranty claims encounter difficulties because the vehicle or battery records cannot be found in the system. This may delay warranty verification, require additional administrative investigation, and affect the efficiency of customer service. An AI-assisted application could help administrators identify possible matching records, detect missing information, and recommend appropriate verification steps. The final decision on warranty eligibility would remain with authorised personnel.

3.0 PROPOSED SYSTEM AND AI APPROACH

To address this challenge, this project proposes an AI-Assisted Battery Warranty Verification and Claim Management System for Bateriku.com. The proposed system will use artificial intelligence to assist administrators in searching for relevant records, matching customer-submitted information with existing vehicle and battery records, identifying incomplete or potentially inconsistent information, and supporting preliminary warranty verification. If no matching record is found, the system can guide the administrator to request supporting documents, such as a purchase receipt or battery serial number, and refer the case for manual verification rather than automatically rejecting the claim.

The proposed system aims to improve the efficiency, consistency, and traceability of warranty claim processing while maintaining human oversight. AI-generated recommendations will support, rather than replace, the decisions of authorised administrators, technicians, and warranty personnel. The system will also support the recording of inspection outcomes and replacement decisions in accordance with the applicable warranty policy. Its effectiveness will be evaluated using appropriate measures, including record-matching accuracy, verification time, missing-record handling, and the rate of incorrect warranty recommendations.

Overall, this project explores how AI can support battery warranty verification and claim management by reducing manual record-searching effort, improving information accessibility, and helping staff handle missing records more systematically. The proposed solution will be evaluated against a non-AI alternative, such as a structured warranty verification checklist and manual record-search process, to determine whether AI provides measurable benefits.

4.0 OBJECTIVE SYSTEM

The objectives of this project are;

i. To investigate the existing battery warranty claim process at Bateriku.com and identify issues related to missing or incomplete customer, vehicle and battery record.
ii. To propose an AI-assisted system that helps administrators match customer claim information with existing warranty records and identify cases requiring further verification.
iii. To evaluate the proposed system using suitable measures, including record-matching accuracy, claim verification time, and the handling of missing or inconsistent information.
iv. To ensure that warranty decisions remain accountable, explainable, and subject to authorised human review.

5.0 SCOPE SYSTEM

The project focuses on designing an AI-Assisted Battery Warranty Verification and Claim Management System for Bateriku.com. The proposed system aims to assist administrators and warranty officers in verifying customer battery warranty claims, particularly when vehicle or battery records are missing, incomplete, or inconsistent in the existing system.

The system will use AI-assisted record matching to identify potential matches between customer-provided information and existing warranty records. It may also extract relevant information from submitted receipts or warranty documents and flag cases that require further investigation. Based on the available evidence, the system will provide recommendations to support staff in deciding the next appropriate action.

The proposed system will support three main outcomes: a potential warranty record is found, additional information is required, or manual verification is needed. Authorised staff will remain responsible for confirming warranty eligibility and making the final decision.

The project will evaluate the proposed system using record-matching accuracy, claim verification time, the handling of missing or inconsistent information, and the rate of incorrect recommendations. The system will be compared with a non-AI approach involving structured checklists and manual warranty record searches.

The project will focus on the warranty verification stage and its supporting claim information. It will not replace physical battery inspections, technician judgement, official warranty policies, or final decisions made by authorised personnel.

6.0 TARGET USER AND STACKHOLDERS

The proposed system involves four main stakeholder groups;

i. Administrator
   - Search and verify warranty status using vehicle registration number or customer phone number.
   - When the status is confirmed, admin will request Bhero/technician go to customer location.

ii. Technician
   - Reviews the original complaint and service summary, performs appropriate inspections and records actual technical findings such as CCA reading and millage reading.

iii. Customer
   - Provides information about vehicle symptoms and receives appropriate service.

iv. HQ Inspector
   - Record inspection result within authorised process.

7.0 REQUIREMENTS AND USER STORIES

7.1 Functional Requirements

The proposed prototype will include the following functional requirements;

i. Search warranty records.
   - The system allows the admin to search for a customer's vehicle information using vehicle registration number or customer phone number.

ii. Suggest possible matching records using AI.
   - If the record cannot be found, the system will suggest possible matching records based on the available information such as battery serial number.

iii. Check brand and battery size.
   - Warranty period different according types of battery. For example, for common batteries the warranty period is 18 months but for EFB and AGM batteries the warranty period is 12 months.

iv. Check warranty period.
   - The system shall display the recorded warranty start date, expiry date and warranty status.

v. Record claim information.

- The system allows the admin to record the customer warranty claim and the verification outcome.

vi. Support human verification

- The system should allow the admin to review AI suggestions and confirm or reject a proposed record match.

7.2 Non-Functional Requirements

Non-functional requirements describe the quality, performance, security and operating constrains of the system.

i. Performance

- The system should display search result within 3 seconds under normal operating conditions.

ii. Security

- Only authorised user shall be allowed to access customer and warranty records.

iii. Reliability

- The system should handle missing or incomplete information.

iv. Usability

- The interface should be user friendly and admin can understand easily.

7.3 User Stories

Story 1:

Administrator

- Trying to match the vehicle registration number in the record. If the record cannot be found, there have alternative way to find the record.

Story 2:

Technician

- Want to verified warranty status either still under warranty or not and have to go to customer location with temporary floating battery if the battery really damages.

Story 3:

HQ Inspector

- Do the second inspection and when the warranty approved, the original warranty date are retained and customer will give a new battery.

8.0 QUALITY ATTRIBUTES SCENARIOS

Quality attribute scenarios define the expected behaviour and measurable quality of the proposed AI-Assisted Battery Warranty Verification System for Bateriku.com under specific conditions. They help ensure that the system is not only functional but also reliable, secure, efficient and easy to use.

8.1 Performance

When an admin searches for a customer's vehicle or battery warranty record, the system should return the search results within an acceptable time under normal operating conditions. A proposed target is to display results within three seconds for at least 95% of searches. This will help admins verify warranty information efficiently and reduce delays in processing customer claims.

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8.2 Reliability

When the system cannot find an exact vehicle or battery record, it should display a clear notification and suggest possible matching records if sufficient information is available. The system must not automatically approve or reject the warranty claim based on an AI suggestion. Cases with missing or conflicting information must be referred to an authorised admin for manual verification. This helps prevent incorrect warranty decisions.

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8.3 Security

The system should protect customer information by allowing access only to authorised users. When an unauthorised user attempts to access warranty records, the system should deny access. Important security events should be recorded where appropriate. These measures help protect customer privacy and reduce the risk of unauthorised disclosure of information.

8.4 Usability

The system should provide a clear and understandable interface for admins. When the AI suggests possible matching records, the system should display relevant details, such as the vehicle registration number, battery serial number and warranty dates when available. The interface should clearly distinguish AI suggestions from verified records so that admins can make informed verification decisions.

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9.0 LIGHTWEIGHT SYSTEM ARCHITECTURE

The proposed AI-Assisted Battery Warranty Verification System for Bateriku.com adopts a lightweight three-layer architecture consisting of the user interface layer, application logic layer and data storage layer. This architecture is selected to keep the prototype simple, modular and suitable for demonstrating the proposed solution to the problem of missing vehicle or battery warranty records.

The **user interface layer** allows an admin to enter available customer, vehicle or battery details and review the search results. For the initial prototype, a simple command-line interface is sufficient to demonstrate the system's functionality. For example, an admin receives a warranty claim from a customer whose car battery may still be under warranty. The admin enters details such as; vehicle registration number, battery serial number and customer phone number.

The **application logic layer** processes the information entered by the admin. It performs exact record searches, checks recorded warranty dates and uses an AI-assisted matching function to suggest possible records when an exact match cannot be found. The prototype uses text-similarity matching rather than a trained machine-learning model. All suggested matches must be reviewed by an authorised admin before being used for warranty verification.

The **data storage layer** contains the vehicle, battery and warranty information required for searching and verification. The initial prototype uses synthetic data to demonstrate the workflow without exposing actual customer information. In a future implementation, the system could connect to an authorised warranty database, subject to appropriate technical access and security controls.

The layers interact when an admin submits a search request. The application logic retrieves the relevant records, evaluates possible matches and returns the results to the user interface. If the exact record is unavailable or the information is inconsistent, the system flags the case for manual review rather than automatically approving or rejecting the warranty claim.

This architecture supports a manageable prototype while separating user interaction, processing logic and data storage. It also supports future improvements, such as a graphical user interface, a more advanced matching model and integration with an authorised warranty database.

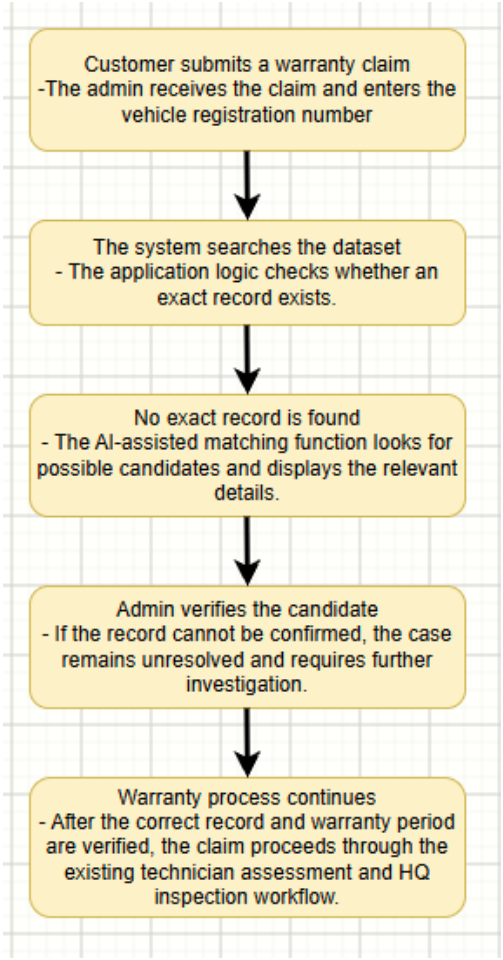


Figure 9.1: AI-Assisted Battery Warranty Verification and Claim Management System Workflow

10.0 INITIAL RISK REGISTER

An initial risk register was prepared to identify potential risks associated with developing the AI-Assisted Battery Warranty Verification System for Bateriku.com. The main risks relate to incorrect AI-generated record matches, limited dataset availability, customer data security, inaccurate warranty information, software defects and limited access to the existing business system.

Risk	Stakeholder	Consequence	Likehood/impact	Control	Owner/evidence
Missing record	Customer/Admin	legitimate claim delayed or refused	Medium / High	"Unable to verify" status; manual investigation; no default rejection	HQ inspector and Admin; test missing/no-match cases
Incorrect warranty date	Admin/Technician	wrong eligibility or replacement handling	Medium / High	Authoritative dates, validation, boundary tests, preserve original dates per policy	Admin and Technician; approved policy review and date tests
Incorrect AI matching	Admin	The AI might suggest a record belonging to a different vehicle because some details are similar.	Medium / High	Display the candidate's details, identify the result as an AI suggestion and require an authorised admin to confirm the match. If the evidence is insufficient, leave the claim unresolved.	Customer and Admin
Customer data exposure	Customer	If real customer information is used without suitable protection, sensitive details could be exposed.	Medium / High	system would need appropriate authentication, access permissions and secure handling of customer information.	Customer

Figure 10.1: Initial risk register for AI-Assisted Battery Warranty Verification and Claim Management System for Bateriku.com

11.0 TRACEABILITY MATRIX

A requirements traceability matrix is used to establish relationships between the identified business problem, supporting evidence, system requirements, architecture components, risks, controls and planned verification activities.

For the proposed AI-Assisted Battery Warranty Verification System for Bateriku.com, the matrix ensures that each important requirement is linked to the problem it addresses and the component responsible for implementing it. It also identifies potential risks, such as incorrect AI-generated record matches, and links these risks to suitable controls, including human verification.

Planned verification activities are included to determine whether the requirements and controls behave as expected. For example, the AI-assisted matching requirement will be tested using synthetic records containing exact matches, partial matches, missing information and unrelated identifiers. The results will be checked to determine whether candidate records are presented appropriately and whether the system requires human verification.

The matrix supports forward and backward traceability. Forward traceability helps demonstrate that identified problems are addressed by requirements and tested features. Backward traceability helps confirm that proposed system features have a clear justification. The matrix will be updated as requirements, architecture decisions, risks and test results change during development.

Problem Evidence	Requirements	Architecture components	Risks	Control	Planned Verification
customer's vehicle record cannot be found.	Search for warranty records using available vehicle or battery details.	Record search module; data storage layer.	Missing or incomplete data.	Display a clear missing-record message.	Search for an existing record and a non-existent record; check both results.
An exact record may be unavailable, Bhero/technician making manual investigation necessary.	Suggest possible matching records using available details.	AI-assisted matching module.	Incorrect detail suggested	Show candidate details and require admin verification.	Test exact, partial, missing and unrelated record details; check the suggestions.

Warranty verification depends on the recorded warranty period.	Display and check warranty dates.	Warranty checking module; data storage layer.	Missing or incorrect dates cause a misleading status	Validate dates	Test valid, expired, missing and inconsistent warranty dates.
The workflow requires the correct record to be identified before proceeding with the claim.	Allow the admin to review and confirm a suggested record.	User interface; human verification function.	Admin accepts an incorrect suggestion.	Clearly label suggestions as unverified and record the admin's decision.	Check that suggestions cannot be treated as verified until the admin confirms them.
The claim proceeds through technician assessment and HQ inspection.	Record the claim status and inspection outcome.	Claim management logic; data storage layer.	Claim information is incomplete or incorrectly updated.	Validate required fields and record status changes.	Test status updates, missing required fields and audit entries.

Figure 11.1: Traceability matrix for AI-Assisted Battery Warranty Verification and Claim Management System for Bateriku.com

12.0 CONCLUSION

In conclusion, the proposed AI-Assisted Battery Warranty Verification and Claim Management System for Bateriku.com aims to improve the efficiency, accuracy, and consistency of the battery warranty claim process. The system will use available warranty records and battery test results to assist administrators and technicians in verifying warranty eligibility and reviewing battery conditions. It may also help identify incomplete or missing records that require further investigation.

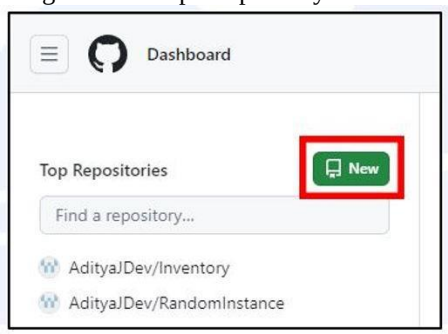
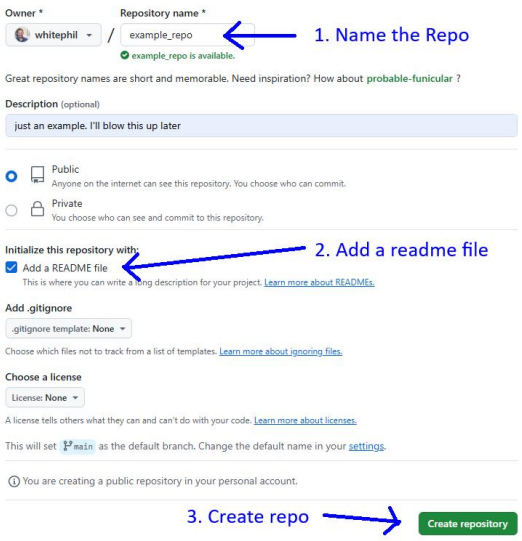
The proposed system is intended to support, rather than replace, human decision-making. Administrators and qualified technicians should remain involved in reviewing uncertain cases and making final decisions, particularly when records are incomplete or inspection results are inconclusive. To promote responsible AI use, the system should consider data privacy, transparency, fairness, reliability, and accountability.

Overall, this project provides an opportunity to explore how AI can support warranty claim management in an automotive service context. Its effectiveness should be evaluated using appropriate data, testing procedures, and feedback from relevant stakeholders. With proper validation and human oversight, the proposed system has the potential to improve the warranty verification process while reducing avoidable errors and supporting more consistent decisions.

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Appendix

Generative AI Usage Declaration

AI Tool Used	Prompts	Outputs	Justifications
ChatGPT by OpenAI	Assist me how to create repository dalam github.	Langkah 1 — Cipta repository baharu  1. Buka GitHub Pergi ke GitHub.com dan log masuk ke akaun awak.  2. Pilih New repository Klik tanda + di bahagian kanan atas GitHub, kemudian pilih New repository. 3. Isi maklumat repository Gunakan cadangan ini: Repository name mci1014-battery-warranty-ai Salin	This output was adopted because it establishes the problem the proposed system aims to address, identifies the users affected, and explains the potential benefits and risks. The problem statement also provides a clear basis for defining the system requirements.

Description

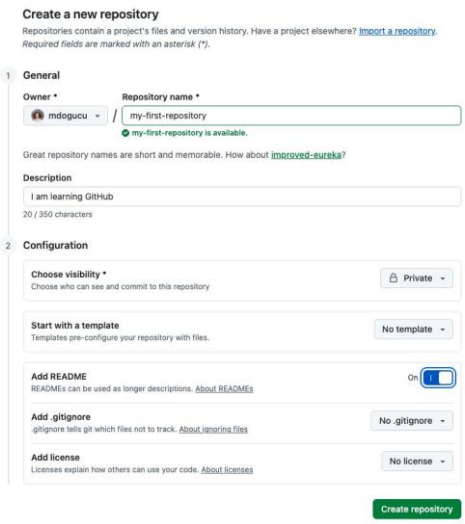
AI-Assisted Battery Warranty Verification and Claim Management System for Bateriku.com

Visibility

Pilih Private jika awak mahu menghadkan akses kepada repository. Pastikan pensyarah boleh mengaksesnya jika tugasan memerlukan perkongsian.

Initialize repository

Tandakan Add a README file.



4. Klik Create repository

Selepas itu, GitHub akan membuka repository baharu awak.

Langkah 2 — Jangan cipta fail lain dahulu

Selepas repository berjaya dibuat, kita akan bina struktur folder yang betul. Contohnya, struktur awal yang mungkin kita gunakan ialah:

```
mci1014-battery-warranty-ai/
├── README.md
├── requirements.txt
├── data/
├── notebooks/
├── src/
├── tests/
├── artifacts/
├── docs/
└── .github/
    └── workflows/
```

		Ini struktur cadangan awal sahaja. Kita akan sesuaikan dengan keperluan sebenar assignment dan rubric awak, bukan sekadar mencipta fail secara rawak. Sekarang, buat Langkah 1 dahulu. Selepas awak klik Create repository, beritahu saya sama ada repository baharu itu sudah berjaya dibuka. Kita akan teruskan langkah seterusnya bersama-sama. ok saya sudah create respository MCI1014-AI-Assisted-Battery-Warranty Bagus! 👍 Sekarang kita sudah ada repository baharu untuk tugasan MCI1014. Kita akan susun fail dengan betul supaya memenuhi keperluan Reproducible Repository dalam assignment. Repository awak bernama: MCI1014-AI-Assisted-Battery-Warranty Kita buat satu langkah pada satu masa. Jangan cipta fail lain dahulu. Langkah 2 — Semak halaman repository Pastikan awak berada di halaman repository baharu di GitHub. Jika awak telah menandakan Add a README file semasa mencipta repository, awak sepatutnya nampak fail README.md.	
ChatGPT by OpenAI	Help identify the stakeholders, their roles, and the decisions supported by the proposed AI-based car battery replacement system.	The stakeholder and decision mapping involving customers, battery technicians, and operations managers, together with the decisions each stakeholder makes or contributes to.	This output was adopted because the system needs to support different stakeholders with different responsibilities. The mapping helps clarify that AI provides decision support while technicians retain responsibility for the final battery replacement decision.

ChatGPT by OpenAI	can you help me to suggest assignment topic using the data given that related in bateriku.com , focus in warranty. the process of warranty at bateriku.com is, customer will make warranty claim, admin will cek in sistem data for that car, if their still in warranty period, technician will go to location, if the battery really problem, so technician will give floating battery for temporary usage to customer. The battery that customer want to do warranty will sent to HQ bateriku.com for inspection. If the battery really rosak than customer will get a new battery dan the warranty period will continue from the first warranty. the problem sometimes happen when there is no	1. Suggested assignment topics; Topic 1 — AI-Assisted Battery Warranty Verification and Claim Management System for Bateriku.com Best overall fit Focuses on helping administrators locate and verify battery purchase records, check warranty eligibility, and manage missing records before sending cases for further review. Topic 2 — More focused AI-Based Missing Battery Record Detection for Warranty Claims at Bateriku.com Focuses specifically on identifying possible matching records when the vehicle registration number, battery serial number, customer details, or purchase information cannot be found. Topic 3 — Broader scope AI-Enabled Warranty Claim Tracking and Temporary Battery Management at Bateriku.com Covers warranty verification, technician visits, temporary floating batteries, shipment to HQ, inspection results, and replacement tracking.	1. Suggested assignment topics; Topic 1 — AI-Assisted Battery Warranty Verification and Claim Management System for Bateriku.com Best overall fit Focuses on helping administrators locate and verify battery purchase records, check warranty eligibility, and manage missing records before sending cases for further review.

<table>
<tr><td></td><td>record about that car in the system. hao AI can be implement for that?</td><td></td><td></td></tr>
<tr><td>ChatGPT by OpenAI</td><td>Can you explain what is functional and non-functional requirement?</td><td>The easiest way to remember is:
Functional requirements = WHAT the system must do.
Non-functional requirements = HOW WELL the system must perform.</td><td>Functional requirements
-Search warranty records.
- Check brand and battery size.</td></tr>
<tr><td>ChatGPT by OpenAI</td><td>What is a quality attribute scenario?</td><td>1. What is a quality attribute?

A quality attribute is a characteristic of a software system that describes its quality.

Examples include:

Performance

How quickly does the system respond?

Security

How well does the system protect customer data?

Reliability

Can the system handle problems without failing?

Usability

Can admins understand and use the system easily?

A quality attribute scenario turns a general quality goal into a specific situation with a measurable expected response.

A commonly used format has six parts:

<table><tr><th>Part</th><th>Meaning</th></tr><tr><td>1. Source</td><td>Who or what causes the event?</td></tr><tr><td>2. Stimulus</td><td>What event happens?</td></tr><tr><td>3. Environment</td><td>Under what conditions does it happen?</td></tr><tr><td>4. Artifact</td><td>Which part of the system is affected?</td></tr></table></td><td>Quality Attribute;
1. Performance
2. Security
3. Reliability
4. Usability</td></tr>
</table>

		<table><tr><td>5. Response</td><td>How should the system react?</td></tr><tr><td>6. Response measure</td><td>How do we measure success?</td></tr></table>	
ChatGPT by OpenAI	Can you explain more details about lightweight system architecture? I'm not familiar with that	1. What is system architecture? Imagine you want to build a house. Before construction begins, you need a plan showing the rooms, doors, electrical wiring and water pipes. Software architecture works in a similar way. It is the plan for your software system. For your Bateriku.com project, the architecture should explain: • Who uses the system? • Where does the warranty information come from? • How does the AI help find missing records? • Where is the information stored? • Who makes the final warranty verification decision? Your system architecture answers these questions before you build the prototype. 2. What does "lightweight" mean? The word lightweight means that the design is kept simple and focuses only on what your assignment needs. You do not need to build a full commercial Bateriku.com system with a mobile application, a cloud server, payment processing, technician GPS tracking and a large AI model. Instead, your prototype can use: • A simple interface or command-line interaction. • A small sample warranty dataset. • A Python program to search for possible matching records. • A human admin to verify the AI suggestions. • Basic tests to demonstrate that the system works. This is enough to demonstrate your proposed solution, provided you clearly explain its limitations.	• A simple interface or command-line interaction. • A small sample warranty dataset. • A Python program to search for possible matching records. • A human admin to verify the AI suggestions. • Basic tests to demonstrate that the system works.
ChatGPT by OpenAI	Can you explain more detail about initial risk register?	Think of it as a plan to answer three questions: 1. What problems might happen when developing or using the system? 2. How likely are those problems, and how much damage could they cause?	1. Admin cannot find a customer's vehicle or battery record in the warranty system.

3. What actions can we take to prevent or manage them?

The word initial means you identify the risks early in the project, before the system is fully developed and tested. You will update the register as you learn more.

1. Why does your project need a risk register?

Your project addresses a specific business problem: sometimes the admin cannot find a customer's vehicle or battery record in the warranty system.

Your proposed AI may help by suggesting possible matching records. However, this introduces new risks. For example, the AI could suggest another customer's battery record by mistake.

If the admin accepts the wrong match, the warranty period might be checked against incorrect information. Therefore, your system needs safeguards.

2. What information does a risk register contain?

A typical risk register contains the following columns.

Column	Meaning	Example
Risk ID	Unique identifier for each risk	R01
Risk description	What could go wrong?	AI suggests the wrong battery record.
Likelihood	How likely is it to happen?	High
Impact	How serious would the consequences be?	High
Risk level	Overall priority based on likelihood and impact	High
Mitigation	What will you do to reduce the risk?	Require admin verification before using a suggested match.
Risk owner	Who is responsible for monitoring the risk?	Project developer or authorised admin
Status	Whether the risk is open, being managed or resolved	Open

You do not need to use exactly these columns unless your lecturer specifies a required format.

Understanding likelihood and impact

2. AI could suggest another customer's battery record by mistake.

<table>
<tr>
<td></td>
<td></td>
<td>

Likelihood means the chance that the risk will occur.
Impact means how serious the consequences would be if it occurs.

For example, an incorrect AI match could have a high impact because it could lead to checking the wrong warranty record. Its likelihood cannot be confirmed without testing, so you would initially estimate it and then revise that estimate using evidence.

A simple scoring method is:

<table>
<tr><th>Score</th><th>Likelihood</th><th>Impact</th></tr>
<tr><td>1</td><td>Very low</td><td>Negligible</td></tr>
<tr><td>2</td><td>Low</td><td>Minor</td></tr>
<tr><td>3</td><td>Medium</td><td>Moderate</td></tr>
<tr><td>4</td><td>High</td><td>Major</td></tr>
<tr><td>5</td><td>Very high</td><td>Severe</td></tr>
</table>
You can calculate a basic risk score as:

\[\text{Risk score}=\text{Likelihood}\times\text{Impact} \]

For example, if likelihood is rated 3 and impact is rated 5, the risk score is \(3 \times 5 = 15\).

One possible classification is 1–4 Low, 5–9 Medium, 10–15 High and 16–25 Very High. These thresholds are an example scoring scheme, not a universal standard.
</td>
<td></td>
</tr>
<tr>
<td>ChatGPT by OpenAI</td>
<td>what is traceability matrix dan how to Connect problem evidence to requirements, architecture components, risks, controls and planned verification.</td>
<td>
The word traceability means being able to follow a decision from its original reason through to its implementation and testing.

This is the connection your assignment is asking you to demonstrate. Every design decision should have a reason, and every important requirement should have a way to verify it.

2. What does each column mean?

For your project, you can use these six columns.

<table>
<tr><th>Column</th><th>What you should write</th></tr>
<tr><td>Problem evidence</td><td>The observation, interview, record or other evidence showing that a problem exists.</td></tr>
<tr><td>Requirement</td><td>What the system must do to address the problem.</td></tr>
<tr><td>Architecture component</td><td>Which part of the system is responsible for implementing the requirement.</td></tr>
</table>
</td>
<td>some vehicle records cannot be found</td>
</tr>
</table>

		Risk	What could go wrong when implementing or using the solution.	
		Control	The safeguard or action used to reduce the risk.	
		Planned verification	The test or review you will perform to check that the requirement and control work as intended.	
		One important distinction: problem evidence is not the same as a risk. Evidence explains what problem has been observed; a risk describes what could go wrong. For example, “some vehicle records cannot be found” is the reported problem. “AI suggests a different customer's record” is a potential risk.		